



Original article

Improved maximum correntropy criterion Kalman filter with adaptive behaviors for INS/UWB fusion positioning algorithm

Yan Wang^{a,b,*}, Shengqing Fu^a, Fuhui Wang^a^a Department of Computer and Communication Engineering, Northeastern University, Qinhuangdao, Hebei Province 066004, China^b Hebei Key Laboratory of Marine Perception Network and Data Processing, Northeastern University, Qinhuangdao, Hebei Province 066004, China

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ABSTRACT

Accurate indoor robot navigation cannot be achieved without reliable indoor positioning techniques. Ultra-wideband (UWB) is among the most dependable techniques currently available. However, the complexity of indoor environments results in signal transmissions that are susceptible to interference from obstacles, which in turn reduces positioning accuracy. Inertial Navigation System (INS) is an autonomous navigation system that is free from interference in indoor environments and unaffected by non-line-of-sight (NLOS) conditions. This paper proposes a new joint INS and UWB positioning method utilizing the Maximum Correntropy Criterion Kalman Filter (MCCKF). This approach effectively cope with the interference of measurement outliers and extend the design of the adaptive mechanisms to enhance the performance of the localization system. For UWB positioning of tag nodes, an improved Particle Swarm Optimization combined with kmeans (PSO-kmeans) method is used to reduce the impact of NLOS errors on positioning. Finally, the INS is calibrated by AMCKKF fused positioning results. The results of simulations and experiments demonstrate that the proposed AMCKKF fusion algorithm effectively suppresses the impact of anomalous measurements, enhances positioning accuracy and robustness, thereby improving its practicality in real-world environments.

1. Introduction

The importance of location information is increasingly evident in industrial production and daily life. In indoor environments, the satellite signals from the Global Navigation Satellite System (GNSS) are relatively weak, making precise positioning challenging. The stability and accuracy of GNSS-based positioning cannot be fully guaranteed [1]. As the demand for precise and reliable indoor robot localization grows, researchers have turned their attention to combinational localization algorithms.

Within the domain of indoor positioning, the signal that is directly transmitted from a tag node to a base station (BS) is termed as Line-of-Sight (LOS) transmission. Nonetheless, in practical applications, signals frequently encounter obstacles, leading to non-Line-of-Sight (NLOS) transmission causing blockage, reflection, or refraction. As the transmission distance in NLOS environments tends to be longer than that in

LOS environments, distance measurements often exhibit significant positive bias, known as NLOS error [2]. This error is a major contributor to inaccurate indoor positioning. As a result, reducing the impact of NLOS error has become a major focus of current indoor localization

research.

Recently, ultra-wideband (UWB) technology has seen extensive use in indoor positioning owing to its superior precision, low transmission power, and excellent penetration capabilities [3]. However, in complex environments, UWB technology can be impacted by multipath propagation and NLOS factors. In contrast, Inertial Navigation Systems (INS) are autonomous navigation systems that do not rely on any external information and acquire measurement data via an Inertial Measurement Unit (IMU). The IMU primarily rely on gyroscopes and accelerometers for measurements. The INS sets up a coordinate framework based on the angular velocity and acceleration data from the gyroscopes and accelerometers, thereby calculating the vehicle's position, velocity, and attitude. Unlike UWB positioning, INS does not require the deployment of beacon nodes, but it does accumulate errors over time due to integration, leading to reduced long-term localization precision. This is a significant drawback of INS. Consequently, the combination of INS and UWB positioning enables the leveraging of complementary advantages, potentially overcoming the individual shortcomings of each system [4].

Kalman filters are classical linear filters, but in practical applications, most kinematic and measurement models are intricate and nonlinear.

* Corresponding author at: Department of Computer and Communication Engineering, Northeastern University, Qinhuangdao, Hebei Province 066004, China
E-mail addresses: wangyan.jgxy@neuq.edu.cn (Y. Wang), 2372391@stu.neu.edu.cn (S. Fu), 2372413@stu.neu.edu.cn (F. Wang).

Despite the widespread utilization of nonlinear filters like the Extended Kalman Filter (EKF) [5], Unscented Kalman Filter (UKF) [6], and Cubature Kalman Filter (CKF) [7] to address such complexities, acquiring accurate noise models remains a daunting task, especially in intricate indoor environments where uncertainties, such as potential anomalies in the UWB localization further exacerbate the model uncertainty. While Kalman filtering commonly employs static noise models for noise estimation, the actual noise characteristics of sensors dynamically evolve over time, potentially degrading filtering performance. The covariance matrices of process noise Q and measurement noise R significantly affect Kalman filter's ability to estimate states accurately. Improper selection of Q and R can notably diminish the effectiveness of the Kalman filter, even leading to filter divergence [8].

The Kalman filter, along with its aforementioned extended algorithms, operate on the basis of the Minimum Mean Square Error (MMSE) criterion, utilizing the second-order statistical characteristics of errors signals. However, their capacity to mitigate non-Gaussian noise is constrained [9]. During the process of integrated navigation position estimation, various factors may lead to the occurrence of anomalous positioning points. Therefore, the introduction of Maximum Correntropy Criterion (MCC) into the Kalman filtering process is proposed to enhance the adaptability of filtering algorithms to non-Gaussian noise, substituting the MMSE criterion.

In the Maximum Correntropy Criterion Kalman Filter (MCCKF), the kernel size linked to the kernel function is critical to the filter's efficiency [10]. However, extant adaptive methods for kernel bandwidth determination are prone to over-convergence. If the kernel bandwidth is excessively oversized, the Kalman filter based on MCC may degrade to a conventional Kalman filter, losing its robustness against significant outliers. Conversely, if the kernel bandwidth is excessively small, the MCCKF might either diverge or exhibit slow convergence.

This paper proposes an improved MCC-KF with adaptive behavior for integrating INS/UWB localization algorithms. By leveraging the strengths of both INS and UWB technology, it combines maximum correlation entropy to reduce the effect of non-Gaussian noise and introduces a novel clustering method to alleviate NLOS ranging errors in UWB. The key contributions are as follows:

- 1) A new Kmeans combined with Particle Swarm Optimization (PSO) algorithm, which utilizes PSO to select initial cluster centers and introduces a new convergence criterion: terminating iterations if the change over three consecutive iterations falls below the specified precision. This enhances PSO's global search capability and computational speed on medium-sized datasets. It is used to preprocess UWB data to remove outliers resulting from noise interference, thereby mitigating the impact of NLOS.
- 2) In complex environments, noise characteristics exhibit time variation. We propose an enhanced adaptive Sage-Husa filtering algorithm, which leverages previous process estimates and recorded information to suppress the impact of outliers. We modify the expressions of covariance matrices R and Q to ensure their positively definite, thereby effectively enhancing the adaptability and stability of localization.
- 3) A novel adaptive bandwidth selection method based on NLOS detection. It addresses the issue of over-convergence caused by outliers and achieves adaptability for specific measurements. Furthermore, we ensure optimal estimation in mild NLOS environments.
- 4) Adaptive maximum correlation entropy is utilized to merge UWB and INS to reduce the effect of non-Gaussian noise and to correct INS using the positioning results. Comparative results demonstrate improved adaptability and robustness to non-Gaussian noise compared to recent methods.

2. Related work

Since UWB and INS can complement each other well, positioning algorithms combining UWB and INS have received much attention and applications in indoor positioning research in recent years. For instance, UWB and INS systems are effectively utilized for tracking pedestrians and tag devices [11]. In [12], a novel loosely coupled INS/UWB pedestrian localization method is presented. This method employs a particle filter for position calculation based on UWB joint state estimation, while zero-velocity updates assist the INS implementation. The resulting positioning accuracy significantly surpasses that of using UWB or INS independently.

Kalman filtering is commonly employed for integrating UWB and INS technologies, but the Kalman algorithm needs a priori information such as accurate dynamic and noise models of the UWB and INS errors. In practical settings, the distribution of noise is usually difficult to determine. Reference [13] proposes a method for compensating NLOS errors. Based on inertial principles, 'virtual inertial points' are constructed using the motion trends of the tags and IMU sensors. Environmental factors are then introduced to compensate for positioning errors in various NLOS environments. [14] proposes a method for correcting UWB observations influenced by NLOS anchors. This approach leverages a priori environmental information to generate a LOS/NLOS map of anchor points, which is then used to select the appropriate LOS anchor at the current location. [15] introduces a tightly coupled fusion algorithm designed for nonlinear environments. This algorithm integrates UWB with simultaneous localization and mapping (SLAM) technology using LiDAR, along with an improved particle filter. The LiDAR measurement data are employed to mitigate NLOS errors in UWB positioning. [16] proposes a hybrid wireless localization technique based on the indoor spatial skeleton, leveraging spatial layout information to enhance positioning accuracy. These methods do not require prior noise information. However, UWB systems may generate outliers in NLOS environments, which cannot be effectively handled by conventional Kalman filters, leading to degradation of system accuracy and even filter divergence.

To address existing challenges, researchers have proposed various adaptive filtering algorithms to enhance integrated localization performance. These algorithms use scaling factors to suppress filter divergence or estimate noise statistical properties. Reference [8] introduces a credibility factor to evaluate inconsistencies in theoretical models, indicating the Kalman filter's reliability. It proposes three credibility factor estimation methods to estimate two unknown noise covariances online and resolves the Sage-Husa method's filter divergence issue. The adaptive Sage-Husa filtering method enables real-time noise correction and estimation, characterized by simplicity and effectiveness. [17] modifies the Sage-Husa algorithm's covariance matrix expression to ensure it is positively definite during filtering and introduces a fading factor to reduce positioning filter divergence. Reference [18] presents a novel adaptive extended Kalman filter method, which uses a noise covariance matrix weighting factor for accurate noise covariance matrices. Reference [3] describes an improved adaptive fusion method for NLOS detection in positioning algorithms. This method first computes the ratio characteristics between UWB measured distances and actual distances across various channels. Subsequently, it utilizes support vector machines (SVM) to identify NLOS conditions, determine the number of LOS base stations, and dynamically apply different optimization algorithms for positioning.

In recent years, optimization standards based on Information Theoretic Learning have garnered significant interest among scholars [9]. The MCC has been utilized as an optimal criterion for the design of the MCC-KF instead of the MMSE, exhibiting superior resilience to impulsive non-Gaussian noise [19]. The MCC considers the second-order moments along with higher-order moments of the error to address the anomalies of the equivalent measurement noise, especially in cases with large outliers. Huber-based filters are also commonly used algorithms for addressing non-Gaussian problems based on the maximum

likelihood method [20]. MCC can fully control incorrect measurements during the estimation process, whereas the Huber method does not process small errors, thus making MCC more effective than Huber. However, incorrect kernel sizes may compromise robustness to outliers and even lead to divergence or slow convergence. To address the limitations of single-kernel correlation entropy, [21] proposed a hybrid correlation entropy criterion that combines two different kernel functions. [10], [22] and [23] introduced several adaptive kernel bandwidth methods that adjust the kernel size in real-time. While current adaptive factor methods can sufficiently maintain the kernel size within a suitable range, persistent interference from outliers may lead to over-convergence, resulting in a continuous reduction of the kernel size. The authors of [24] proposed a multi-kernel adaptive kernel size method, but it does not guarantee the optimal kernel size. In [25], the authors presented an improved method combining PSO and Kmeans algorithms to boost the clustering performance of K-means while maintaining its simplicity and efficiency.

3. Proposed method

This section introduces the proposed adaptive fusion localization algorithm framework and its key components, including the combined UWB and INS system as depicted in Fig. 1. In the INS component, the tag node carries an INS device that acquires acceleration and angular velocity information via an Inertial Measurement Unit (IMU). The INS kinematic model is then employed to estimate tag's attitude, velocity, position, and dynamic biases for subsequent localization.

In the UWB component, we employ Time of Arrival (TOA) technology to measure the distance between the tag and the base stations (BS). Initially, the distance data is preprocessed using a combination of PSO and the K-means algorithm to mitigate measurement noise from the BS. Subsequently, the processed data is utilized in a trilateration method, where M measurements are divided into N distinct groups. Each group uses least-squares estimation to obtain the corresponding position estimates and performs NLOS detection.

In the fusion update step, the difference between INS and UWB position estimates is computed. The noise covariance matrix is adjusted by the improved adaptive Sage-husa filter. Finally, the AMCKKF fuses INS and UWB data to obtain the ultimate localization result, correcting the dynamic bias of the inertial sensors.

3.1. UWB localization

3.1.1. Signal model

The UWB localization algorithm utilizes the TOA based method, which requires at least three UWB anchor points to be deployed. The distance between the UWB tag and the UWB base stations is calculated using the round trip time ranging method. The position information of the UWB tag is then determined based on the geometric relationship.

There is one tag node $P_{MN} = (p_x, p_y, p_z)^T$ and the anchor coordinates are $P_{BS,i}^i, i = 1, 2, \dots, M$. This study employs four anchors and one tag to examine the distance between tag node and the BSs as described below:

$$d_i = \|P_{MN} - P_{BS}^i\| \quad (1)$$

The ranging model between tag node and BSs is:

$$d_i(l) = \begin{cases} d_i + w_{LOS} & \text{if LOS} \\ d_i + w_{LOS} + w_{NLOS} & \text{if NLOS} \end{cases} \quad (2)$$

where, d_i and $d_i(l)$ represent the Euclidean distance and measured distance between tag and the i -th BS, respectively, at time l . w_{LOS} is the sensor error and we assume it follows the Gaussian distribution $N(0, 0.1^2)$. w_{NLOS} refers to the error of NLOS caused by obstacle occlusion, which may follow different distribution with the change of environment. P_{MN} is the coordinate of the tag.

3.1.2. NLOS Weakening by PSO k-means algorithm

In indoor environments, NLOS situations are very common. If the UWB system incorporates a substantial NLOS error, it could result in significant positioning inaccuracies in the ensuing computations. Consequently, we opt for the PSO-kmeans algorithm to pre-process the UWB measurement data to mitigate the NLOS error and diminish its influence. The historical UWB measurements from the L measurement windows are divided into two categories over a certain period of time, and the clustering centers of the two categories are calculated. Then, the clustering center with the smaller value is selected as the average of the LOS distance measurements.

K-means clustering is an algorithm that partitions a dataset into a predetermined number of K clusters grounded on distance. The algorithm iteratively classifies each point in the data as the closest center of mass and recalculates the center of mass until it no longer changes. The K-means clustering employs Euclidean distance to group data by measuring the distance between two vectors within the same space.

The selection of the initial clustering center significantly impacts the clustering results of the k-means algorithm. Different initial clustering

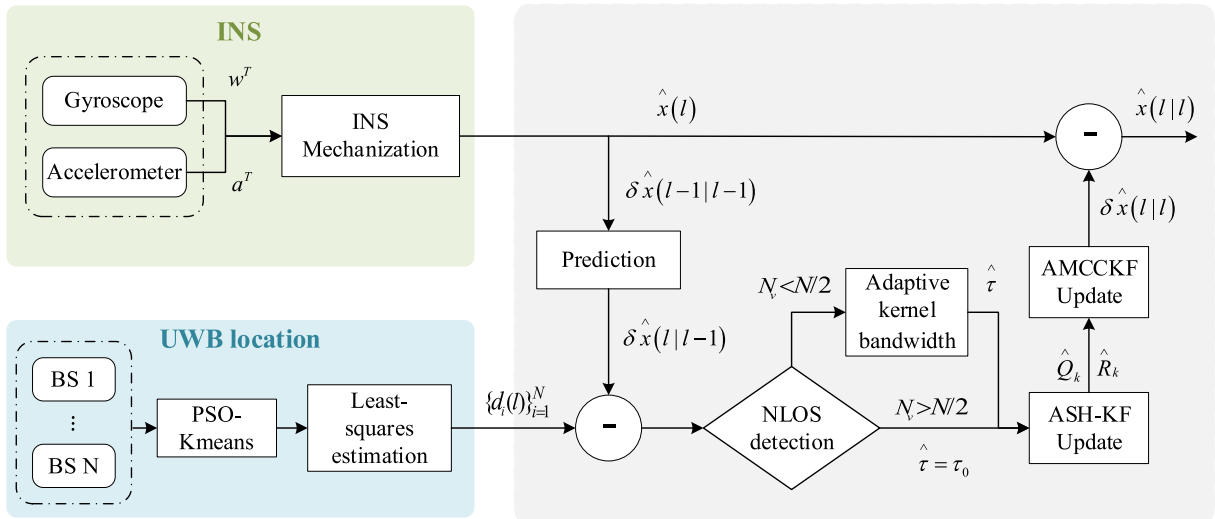


Fig. 1. The framework of the proposed algorithm.

centers may produce varying clustering results. If the initial clustering center selection is unreasonable, the k-means algorithm may be prone to converging upon a local optimal solution. The integration of PSO with the k-means not only prevents the emergence of empty classes, but also has very good global convergence and local optimization capabilities, as well as good results in handling isolated point problems.

The K-means algorithm is characterized by high degree of similarity among objects within the intra-cluster and low similarity among objects in inter-cluster. The PSO algorithm can effectively optimize the initial choice of plasmoids in K-means clustering by randomly generating the initial points and updating these plasmoids in iterations to find the best solution.

The position and velocity of each particle are iteratively revised through the application of specific equations:

$$v_i^{k+1} = w_i v_i + c_1 r_1 (p_i^k - x_i^k) + c_2 r_2 (g_i^k - x_i^k) \quad (3)$$

$$x_j^{k+1} = x_j^k + v_j^{k+1} \quad (4)$$

where, c_1 and c_2 are acceleration factors, and w represents the inertia weight. The parameters for PSO are configured as: $c_1 = 1.5, c_2 = 1.5$ and $w = 0.7$.

The PSO algorithm utilizes a fitness function to identify the optimal solution, whereas K-means assesses clustering quality by minimizing the sum of intra-cluster distances. Therefore, the fitness function can be set as:

$$fitness = \sum_{j=1}^K \sum_{s_i \in C_j} d(s_i, C_j)^2 \quad (5)$$

where, $d(s_i, C_j)$ represents the Euclidean distance, C_j denotes the j -th cluster.

The algorithm contains a new convergence condition. If the change between three generations is less than the allowed precision, i.e., there is almost no change in three updates, the iteration stops and the $flag = 1$. This makes full use of the global search capability of PSO in small and medium sized datasets and improves the computational speed. The PSO-kmeans process can be summarized in the following steps:

Step 1: Begin by initializing the number of cluster categories K , and randomly initialize the populations and velocities of the particles. Each particle signifies a set of K clustering centers. The input consists of UWB measurements, $D = \{d_1, d_2, \dots, d_L\}$.

Step 2: Compute the fitness value for the initial population, identify the initial particle and global optimal solutions.

Step 3: Determine whether the maximum iteration count has been achieved or the convergence condition $flag = 1$ is satisfied, then the output result is used as the initialization of the gbest for the PSO algorithm. If not, update the particle's velocity and position, compute the fitness value, and re-judge.

Step 4: The PSO algorithm's cluster centroids are used as initial points for k-means. The distance between each data point in the dataset and the Kcluster centroids is computed. Subsequently, each data object is then allocated to its closest cluster.

Step 5: Within each category, the cumulative distances from each data point to every other point is computed. The data point with the minimum total distance is chosen as the new center.

Step 6: Redo steps 4 and 5 until the center points not change. Fig. 2 shows the PSO K-means algorithm.

3.1.3. UWB localization

After processing the distance measurements with the clustering algorithm, the impact of NLOS is reduced. The location of tag can be obtained through trilateration using clustered distance measurements. Assume there are M base stations surrounding the tag to be located. At the time k , three of these base stations are selected for positioning. The measurements are then divided into $N = C_M^3$ subgroups, each of which

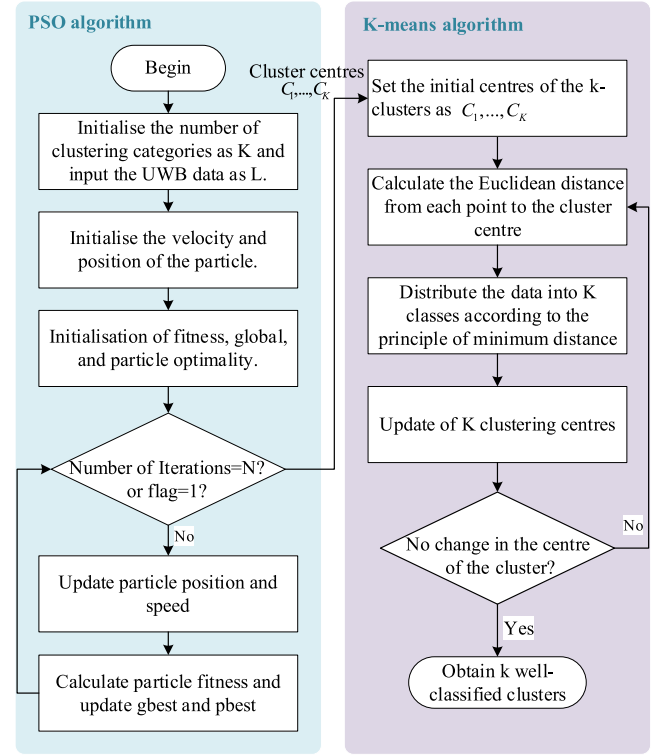


Fig. 2. The framework of the PSO k-means algorithm.

can be resolved using the least squares method. To simplify the operation, we simplify the 3D localization model to a 2D plane, i.e., $z(k) = z_{n,i} = 2$.

$$\begin{cases} (x_1^n - p_x^k)^2 + (y_1^n - p_y^k)^2 = d_{n,1}^2(k) \\ (x_2^n - p_x^k)^2 + (y_2^n - p_y^k)^2 = d_{n,2}^2(k) \\ (x_3^n - p_x^k)^2 + (y_3^n - p_y^k)^2 = d_{n,3}^2(k) \end{cases} \quad (6)$$

where, $n = 1, 2, \dots, N, (p_x, p_y)$ is tag's position at moment k , and $(x_1^n, y_1^n), (x_2^n, y_2^n), (x_3^n, y_3^n)$ are the positions of the three BSs in subgroup n . Eq. (6) can be rewritten as:

$$A = 2 \begin{bmatrix} (x_1^n - x_2^n), (y_1^n - y_2^n) \\ (x_1^n - x_3^n), (y_1^n - y_3^n) \\ (x_2^n - x_3^n), (y_2^n - y_3^n) \end{bmatrix} \quad (7)$$

$$B = \begin{bmatrix} d_{n,2}^2(k) - d_{n,1}^2(k) + x_1^{n2} + y_1^{n2} - x_2^{n2} - y_2^{n2} \\ d_{n,3}^2(k) - d_{n,1}^2(k) + x_1^{n2} + y_1^{n2} - x_3^{n2} - y_3^{n2} \\ d_{n,3}^2(k) - d_{n,2}^2(k) + x_2^{n2} + y_2^{n2} - x_3^{n2} - y_3^{n2} \end{bmatrix} \quad (8)$$

Using the equations provided, the tag's position can be estimated at the given moment k .

$$\begin{bmatrix} p_x^k, p_y^k \end{bmatrix}^T = (A^T A)^{-1} A^T B \quad (9)$$

3.2. INS localization

INS is an external reference independent positioning and navigation technology that infers the position, velocity and attitude of a tag node from measurements of internal sensors. It is adaptable to a variety of environments and can complement well with UWB. In inertial naviga-

tion systems, the motion variations of a tag node are usually described using a three-dimensional orthogonal reference frame, known as the North-East-Down (NED) coordinate framework. The attitude of tag node is denoted by Eulerian angle attitude vectors, $\phi = [\varphi \theta \psi]^T$, φ is roll, θ is pitch, ψ is yaw. The velocity vector of the tag is $\nu = [\nu_n \nu_e \nu_d]^T$, and position vector of the tag is denoted as $p = [p_n p_e p_d]^T$, n, e and d represent the directions of the navigation coordinate system. The state transfer equation of a tag node describes how attitude, velocity, and position change over time. It is based on the data measured by MEMS accelerometers and gyroscopes, and it allows for the calculation of incremental changes in attitude, velocity, and position. The state vector is described in the following:

$$x(t) = [\phi^T \nu^T p^T b_g^T b_a^T]^T \quad (10)$$

where, b_a^T denotes accelerometer bias, b_g^T denotes gyroscope bias. The INS's motion model is described as follows:

$$\dot{x}(t) = [\phi^T \nu^T p^T b_g^T b_a^T]^T = \begin{bmatrix} \frac{1}{2} q^T \otimes (w^T - b_g^T) \\ C_b^n (a^T - b^T) + g^T \\ \nu^T \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

where q^T is tag's attitude quaternion, $\otimes$ represents the quaternion multiplication, C_b^n denotes the direction cosine matrix from the body frame (the reference frame fixed on the tag node) to the navigation coordinate framework, w^T and a^T represents the measurements of the gyroscope and accelerometer respectively, and g denotes the acceleration of gravity within the navigation coordinate framework. By linearizing the system dynamics, the error state equation can be transformed into a set of differential equations

$$\Delta \hat{x}(t) = \begin{bmatrix} \Delta \phi^T \\ \Delta \nu^T \\ \Delta p^T \\ \Delta b_g^T \\ \Delta b_a^T \end{bmatrix} = \begin{bmatrix} -[\omega^T - b_g^T]_{\times} \Delta \phi^T - \omega_{\omega} \\ -C_b^n [a^T - b_a^T]_{\times} \Delta \phi^T - C_b^n \Delta b_a^T \\ \Delta \nu \\ \omega_{\omega} \\ a_{\omega} \end{bmatrix} \quad (12)$$

where, $\Delta x^\wedge(t)$ represents the error state vector, $\Delta \phi^T$ denotes attitude errors, $\Delta \nu^T$ and Δp^T denotes velocity and position errors respectively, Δb_g^T and Δb_a^T denotes the error of gyroscope and accelerometer respectively, $[\cdot]_{\times}$ is skew-symmetrical operations, ω_{ω} denotes the random walk noise of gyroscope, a_{ω} denotes the random walk noise of accelerometer. The system is corrected and adjusted by the error state vector through the use of fused localization feedback from the previous moment.

The error state vector is used for AMCKKF prediction (in Section 6.1.1), which is a small part of INS/UWB integration.

3.3. Adaptive maximum correntropy criterion Kalman filter (AMCKKF)

In practice, in complex indoor environments, INS/UWB integrated positioning is often interfered by time-varying noise, which may result in outliers, leading to scattering of the correlation entropy filters. Sage-Husa is a suboptimal estimation filtering algorithm that theoretically allows adaptive estimation based on system and measurement noise characteristics, and solves the problem of the KF uncertainty in the noise estimation. However, the noise covariance matrix might become non-positive semi-definite, leading to filter divergence. Therefore, we propose a modified adaptive algorithm that utilizes the estimation and

information from the previous process to enhance the regulation of the noise covariance matrix estimation, thus improving the precision of the process. The method not only modifies the expression of the covariance matrix to guarantee that it is positively definite, but also reduces the dispersion of the localization filter by introducing an attenuation factor, which effectively improves the accuracy and stability of localization.

4. Adaptive Sage-Husa Kalman filtering

Adaptive Kalman filtering algorithm is used when the statistical attributes of the system noise are indeterminate. Sage-Husa adaptive Kalman filtering is a classic approach for estimating the covariance of imprecise or unidentified measurement noise using a variance-matching approach, where the covariance matrix is adjusted according to the theoretical value of the innovations or residuals. The innovation value ν_n^l is defined as follows:

$$\nu_n^l = z_n^l - H \delta \dot{X}_{n|n-1} \quad (13)$$

By averaging the approximation and taking the variance on both sides of (13), we receive the covariance:

$$\Gamma_l = \frac{1}{\sigma} \sum_{i=l-\sigma+1}^l \nu_i \nu_i^T = H P_{l|l-1} H^T + R_l \quad (14)$$

where Γ_l is the innovation covariance matrix, σ denotes the size of estimation window. The window size σ has a significant effect on the precision and stability of the estimation. The smaller σ reduces computational complexity and makes the estimation process more adaptive to change, but may lead to filter divergence. Conversely, larger σ improves stability and produces smoother results, but increases computational complexity and may reduce adaptivity. In simulations and experiments, through iterative trials, we set σ to 20 to take these issues into account.

The measurement noise covariance matrix is updated as shown below:

$$R_l = \Gamma_l - H P_{l|l-1} H^T \quad (15)$$

Theoretically, R_l is the covariance matrix, which should be positively definite, however, measurement errors or other factors may result in the estimated covariance matrix that is not strictly positive definite, meaning there may be some negative eigenvalues. In order to keep R_l a positive definite matrix, we use a residual-based approach [22] to estimate R_l :

$$R_l = \Gamma_l + H P_{l|l} H^T \quad (16)$$

Then, from the time-varying noise statistical estimator, $R_l^\wedge$:

$$\hat{R}_l = (1 - d_l) \hat{R}_{l-1} + d_l (\Gamma_l + H P_{l|l} H^T) \quad (17)$$

Where $d_l = \frac{1-b}{1-b^{l+1}}$, $0 < b < 1$ is forgetting factor, typically between 0.95 and 0.99.

To estimate the Q_l , the state prediction noise at time k may be estimated as below:

$$\begin{aligned} \hat{w}_{l-1} &= \hat{x}_l^T - \hat{x}_l^T = K_l (z_n^l - H \delta \dot{X}_{n|n-1}) \\ &= K_l \nu_n^l \end{aligned} \quad (18)$$

Accordingly, the process noise covariance matrix Q_l is approximated as follows:

$$\hat{Q}_l = K_l E [\nu_l \nu_l^T] K_l^T = K_l \Gamma_l K_l^T \quad (19)$$

We can use a similar strategy to $R_l^\wedge$, where $Q_l^\wedge$ can be obtained from a time-varying noise statistical estimator

$$\hat{Q}_l = (1 - a_l) \hat{Q}_{l-1} + a_l (K_l \Gamma_l K_l^T) \quad (20)$$

where, $a_l = \frac{a_{l-1}}{a_{l-1} + b_a}$, denotes the adjustment factor for the Q matrix, b_a akin to the forgetting factor, which enhances the adaptability and tracking capability of the estimates, primarily reliant on current information, thereby reinforcing the significance of residuals.

At each correction step, adaptive estimation and updating of the measurement noise covariance matrix $R^{\wedge}_l$ and the process noise matrix $Q^{\wedge}_l$ are performed using (17) and (20), respectively.

5. Maximum correntropy criterion Kalman filter

The MCCKF is a Kalman filtering algorithm for non-Gaussian measurement noise. The algorithm estimates parameters based on maximizing inter-sample correlation and is better suited to handle outliers and non-Gaussian noise than the traditional MMSE. By utilizing the MCC, the MCCKF can maximize the similarity between two stochastic processes by establishing the maximum similarity between the mathematical model's output and the intended response during the filtering process.

The correntropy is employed to quantify the degree of similarity between two random variables, $X, Y \in R$. The joint distribution function of these variables is given by $f_{XY}(x, y)$. The correntropy is defined as follows:

$$V(X, Y) = E[\kappa_\tau(X, Y)] = \int \kappa_\tau(X, Y) df_{XY}(x, y) \quad (21)$$

where, $\kappa_\tau(\cdot)$ denotes the shift-invariant kernel, which is in accordance with Mercer's theorem, while $E[\cdot]$ represents the expectation operation. The kernel function is chosen to satisfy the continuous positive definite condition, with the Gaussian kernel being a widely used kernel function as follows :

$$\kappa_\tau(X, Y) = G_\tau(e) = \exp\left(-\frac{e^2}{2\tau^2}\right) \quad (22)$$

where, $e = x - y$, τ is the parameter for the kernel size and $\tau > 0$. The equation demonstrates that the correlation function under the Gaussian kernel is non-negative and attains its maximum value when x equals y .

In practice, the $f_{XY}(x, y)$ is typically not known. The correntropy is estimated using N sample means.

$$\hat{V}(X, Y) = \frac{1}{N} \sum_{i=1}^N G(e_i) \quad (23)$$

where, $e_i = x_i - y_i$, and $(x_i, y_i)_{i=1}^N$ represents N sampled data from $f_{XY}(x, y)$. The above Gaussian kernel function undergoes a Taylor series expansion, namely as follows

$$V(X, Y) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n \tau^{2n} n!} E[(X - Y)^{2n}] \quad (24)$$

As can be seen from Eq. (24), the correntropy constitutes a weighted summation of all even-order terms of the error variable $(X - Y)$. Compared to the MMSE, this demonstrates the capacity of correntropy to encompass higher-order terms. Thus, the MCC's cost function is presented below:

$$J_{MCC} = \frac{1}{N} \sum_{i=1}^N G_\tau(e_i) \quad (25)$$

The kernel bandwidth τ serves as the correntropy parameter for terms of second order and above, and the choice of an appropriate τ significantly impacts the filter's performance. When the signal is non-Gaussian noise, the MCC criterion demonstrates its advantage in capturing higher-order terms and suppressing non-Gaussian noise. However, if the kernel

size is substantial, the second-order term dominates, leading to the degradation of MCC to MMSE filtering. Therefore, we propose a method for adapting the size of the kernel bandwidth.

6. Adaptive kernel bandwidth size selection

Covariance matching is a well-known adaptive estimation method that adjusts the covariance matrix based on the theoretical values of the innovations or residuals. The size of the adaptive kernel bandwidth is associated with the measurement noise, and each anchor point may have a different noise distribution. However, this method does not guarantee an optimal bandwidth under normal conditions, so we propose an adaptive bandwidth selection method for NLOS detection that guarantees both adaptive bandwidth handling of outliers and optimal estimation in a mild NLOS environment.

NLOS detection:

Firstly, the M base stations are partitioned into $N = C_M^3$ distinct localization sub-groups, with every three being a group, with index sets $\{n | n = 1, 2, \dots, N\}$. The difference between the measured and predicted values of M base stations and tag nodes is termed as 'innovation':

$$v_n(l) = z_n(l) - \hat{z}(l | l - 1) \quad (26)$$

Assume if none of the measurements in the n-subset contain NLOS errors:

$$v_n(l) \sim N(0, S_n(l)) \quad (27)$$

where, $S_n(l)$ represents the innovation covariance matrix, and can be calculated as follows:

$$S_n(l) = H_n(l) P(l | l - 1) H_n^T(l) + R_n \quad (28)$$

To verify the above equation, we propose a hypothesis test:

$$\begin{aligned} H_{0,n} : v_n(l) &\sim N(0, S_n(l)), n = 1, 2, \dots, N \\ H_{1,n} : &\text{not } H_{0,n} \end{aligned} \quad (29)$$

In the event that $H_{0,n}$ is true, it is assumed that all BSs within the n-th subset are in LOS; conversely, in the event that $H_{1,n}$ is true, it is assumed that at least one subset exists in a NLOS environment, in which case the subset has a significant error in its position estimation. The test statistic is calculated as stated below:

$$T_n(l) = v_n^T(l) S_n(l) v_n(l) \sim \chi^2(2), n = 1, 2, \dots, N \quad (30)$$

Compare the test statistic value $T_n(l)$ to the validation gate threshold value γ . If $T_n(l) < \gamma$, then the subgroup does not contain NLOS error measurements, $H_{0,n}$ holds and the test passes. Otherwise, $H_{1,n}$ holds and the test fails. The γ is related to its probability P_g and the threshold γ gets determined as follows:

$$P_g = 1 - P_{FA} = \int_0^\gamma f_{\chi^2(2)}(x) dx \quad (31)$$

where, P_{FA} denotes the predefined false alarm rate and $f_{\chi^2(2)}(x)$ denotes the chi-square distribution with two degrees of freedom. When the position estimation falls into the estimated probability of the verifiers is not less than 0.99, the chi-square distribution threshold value $\gamma = 9.21$ with a degrees of freedom of 2.

Record the number of localization subset of N subgroups that make $H_{0,n}$ hold, denoted as N_v . By the number of N_v , we can classify the NLOS situation in the localization environment into mild and severe NLOS scenarios. A larger N_v indicates a greater number of subsets with a measurement result in the LOS situation, which indicates the degree of non-visual distance is less severe. Conversely, the smaller N_v is, the fewer subgroups pass the test, which indicates the more serious the degree of non-visual distance in the localization environment.

Adaptive kernel bandwidth scheme:

We define that if $N_v > N/2$, the majority of the subsets pass detec-

tion, while a small fraction of the other subsets do not, and there are measurements with NLOS errors in these subsets that do not pass detection. Although some of all the subsets are contaminated by NLOS, the NLOS severity is not particularly high as some subsets still pass the detection, and this situation is classified as mild situation. In this case, we use a fixed optimal bandwidth, however, if the non-visual distance is severe for $N_v \leq N/2$, we use an adaptive bandwidth.

The adaptive bandwidth τ is adjusted to accommodate different types of noise. Firstly, the relationship between the innovations at each moment and its covariance matrix $S(l)$ is used to identify the existence of measurement anomalies. Define the innovation of KF-MCKF at the l th moment as:

$$v(l) = z(l) - z(l|l-1) \quad (32)$$

In the ideal case, the observation noise is considered to follow a zero-mean Gaussian distribution. Therefore, according to the theory of Kalman filter, the state estimation of Kalman filter should be unbiased, meaning that the expected value of the innovation value is zero. The covariance $S(l)$ of the innovation $v(l)$ is defined as follows:

$$S(l) = H(l)P(l|l-1)H^T(l) + R_l \quad (33)$$

Under the Gaussian assumption, the innovation $v(l)$ should follow a Gaussian distribution with zero-mean and the covariance of $S(l)$, so the squared innovation's Mahalanobis distance should be chi-square distributed with the dimension of the observation as its degree of freedom.

$$u_l = v(l)^T S(l)^{-1} v(l) \sim \chi^2(2) \quad (34)$$

Therefore, the detection statistic in (29) can be used as a condition for judging that, in the absence of outliers, the filter is able to converge strictly and its theoretical covariance aligns with the actual value. However, when outliers are present, the kernel width τ can be dynamically modified and the upper limit of the MCCKF can be satisfied, thereby ensuring that the filter outperforms the CKF in terms of estimation accuracy.

$$\hat{\tau}_j = \begin{cases} \tau_{\max} / \sqrt{\mu_l} & \text{if } N_v \leq N/2 \\ \tau_0 & \text{otherwise} \end{cases} \quad (35)$$

where $\tau_{\max}$ denotes the preset maximum value of the kernel bandwidth, which will be corrected by the adaptive factor, and τ_0 is the optimal bandwidth under normal conditions. The kernel bandwidth will be dynamically modified in real time based on changes in the innovation value. When anomalous measurements are present, the innovation value tends to increase, resulting in a decrease in the kernel bandwidth in Eq. (35), thus suppressing the effect of the outliers. When the NLOS case is mild, the τ optimum of MCCKF can be obtained quickly. Thus the proposed algorithm is both robust and efficient in choosing the kernel bandwidth size.

6.1. INS/UWB fusion based on AMCKF

6.1.1. Improved Sage-Husa Kalman filtering

INS error state propagation is used for AMCKF prediction. The following presents the predicted error state along with its corresponding prediction covariance:

$$\hat{\delta x}(l|l-1) = \Phi_{l|l-1}(\hat{\delta x}(l-1|l-1), u_{l|l-1}) + \zeta \quad (36)$$

$$P(l|l-1) = \Phi_{l|l-1}P(l-1|l-1)\Phi_{l|l-1}^T + Q_{l|l-1} \quad (37)$$

where, $\Phi_{l|l-1} = I + F_{l|l-1}\delta t$ represents discrete-time state transition matrix, $u_{l|l-1} = (\omega^T(l-1), a^T(l-1))^T$ represents system input vector, ζ is system white noise with covariance matrix Q , and $Q_{l|l-1} = G(l|l-1)QG(l|l-1)^T\delta t$, F is continuous-time state transition matrix as

formulated in (38), G is control matrix as formulated in (10), and δt is IMU sampling time period.

$$F = \begin{bmatrix} C_b^n[\omega^T(l-1) - b_\omega^T]_\times & 0 & 0 & C_b^n & 0 \\ C_b^n[a^T(l-1) - b_a^T]_\times & 0 & 0 & 0 & -C_b^n \\ 0 & I_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{\tau_\omega} & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{\tau_a} \end{bmatrix} \quad (38)$$

$$G = \begin{bmatrix} C_b^n(l|l-1) & 0 & 0 & 0 \\ 0 & -C_b^n(l|l-1) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & I_3 \end{bmatrix} \quad (39)$$

where, $C_b^n(l|l-1)$ represents the updated DCM computed from the most recent attitude update. τ_ω, τ_a represent the gyroscope and accelerometer correlation times, respectively. The observation model and innovation are given by:

$$z_n(l) = P_{INS}(l) - P_{UWB}(l) \quad (40)$$

$$z(l|l-1) = H\hat{\delta x}(l|l-1) \quad (41)$$

$$v_n(l) = z_n(l) - z(l|l-1) \quad (42)$$

$$S(l) = HP(l|l-1)H + R_l \quad (43)$$

where, $z_n(l)$ is the error estimate observation vector, $z(l|l-1)$ is the error estimate prediction value, R_l is the covariance matrix of error observation value, $v_n(l)$ is the innovation, $S(l)$ denotes the innovation covariance matrix, and H denotes the observation matrix shown below as:

$$H = [0 \quad 0 \quad I_3 \quad 0 \quad 0] \quad (44)$$

As previously described in (30), the $S(l)$ is used to calculate the test statistic $T_n(l)$.

$$T_n(l) = v_n^T(l)S(l)v_n(l), n = 1, 2, \dots, N \quad (45)$$

If $T_n(l) < \gamma$, we record the count of $z_n(l)$ passing through validation threshold as N_v . Based on the value of N_v , we categorize it as either mild NLOS contamination or heavy NLOS. When $N_v \geq N/2$, it indicates that the position estimate falls within the safety gate, suggesting mild NLOS contamination. In this case, we use the optimal kernel bandwidth τ obtained from multiple experiments. Otherwise, we use the adaptive kernel bandwidth (35).

Then, the Γ_l obtained under the sliding window (mean) is calculated and the measurement update is performed to obtain the improved positive definite $R^{\wedge}_l$ and $Q^{\wedge}_l$.

$$\Gamma_l = \frac{1}{\sigma} \sum_{i=k-\sigma+1}^l \nu_i \nu_i^T \quad (46)$$

$$\hat{R}_l = (1 - d_l)\hat{R}_{l-1} + d_l(\Gamma_l + HP_{l|l}H^T) \quad (47)$$

$$K_l = P_{l|l-1}H^T(HP_{l|l-1}H^T + R_l)^{-1} \quad (48)$$

$$\hat{Q}_l = (1 - a_l)\hat{Q}_{l-1} + a_l(K_l\Gamma_l K_l^T) \quad (49)$$

6.1.2. Improved AMCKF update

From the measurement equations, we can deduce the system model

as follows:

$$\begin{bmatrix} \delta\hat{\mathbf{x}}(l|l-1) \\ \mathbf{z}(l) \end{bmatrix} = \begin{bmatrix} \mathbf{x}(l) \\ H(\mathbf{x}(l)^m, \mathbf{x}(l)) \end{bmatrix} + \boldsymbol{\varpi}(l) \quad (50)$$

where, $\boldsymbol{\varpi}(l) = \begin{bmatrix} \delta\hat{\mathbf{x}}(l|l-1) - \mathbf{x}(l) \\ \delta(l) \end{bmatrix}$ and $\delta(l)$ denotes the measurement noise. Then, we receive the covariance:

$$\begin{aligned} E[\boldsymbol{\varpi}(l)\boldsymbol{\varpi}(l)^T] &= \begin{bmatrix} P(l|l-1) & 0 \\ 0 & R_l \end{bmatrix} \\ &= \begin{bmatrix} B_p(l|l-1)B_p^T(l|l-1) & 0 \\ 0 & B_r(l)B_r^T(l) \end{bmatrix} \\ &= B(l)B^T(l) \end{aligned} \quad (51)$$

where, $B(l)$ is derived from the Cholesky decomposition of $E[\boldsymbol{\varpi}(l)\boldsymbol{\varpi}(l)^T]$, $R_l = R^{\wedge}_l$ is positive definite by (17).

Simultaneous multiplication of the both sides of Eq. (51) by $B^{-1}(l)$ gives $D(l)$:

$$D(l) = W(l)\delta\hat{\mathbf{x}}(l|l-1) + e(l) \quad (52)$$

$$\text{where, } D(l) = B^{-1}(l) \begin{bmatrix} \delta\hat{\mathbf{x}}(l|l-1) \\ \mathbf{z}(l) \end{bmatrix} \quad (53)$$

$$W(l) = B^{-1}(l) \begin{bmatrix} \mathbf{x}(l) \\ H(\mathbf{x}(l)^m, \mathbf{x}(l)) \end{bmatrix} \quad (54)$$

$$e(l) = D(l) - W(l)\delta\hat{\mathbf{x}}(l|l-1) \quad (55)$$

Then, we use MCC to define the cost function.

$$J_{MCC}(\mathbf{x}) = \sum_{j=1}^{n+m} G_{\tau}(e_j(l)) = \sum_{j=1}^{n+m} G_{\tau}(d_j(l) - w_j(l)) \quad (56)$$

where, $e_j(l)$ is the j -th element of $e(l)$, $d_j(l)$ is the j -th element of $D(l)$, $w_j(l)$ corresponds to the j -th row of $W(l)$, and a dimension of $D(l)$ is $L = n + m$.

The optimal estimate of $\delta\hat{\mathbf{x}}(l|l-1)$ under the MCC is obtained through the maximization of the cost function.

$$\begin{aligned} \delta\hat{\mathbf{x}}(l|l-1) &= \operatorname{argmax}_{\mathbf{x}} J_{MCC}(\mathbf{x}) \\ &= \operatorname{argmax}_{\mathbf{x}} \sum_{j=1}^L G_{\tau}(e_j(l)) \end{aligned} \quad (57)$$

By taking a partial derivative of the cost function, we get:

$$\frac{\partial J_{MCC}(\mathbf{x})}{\partial \mathbf{x}(l|l)} = 0 \quad (58)$$

$$\text{namely, } \sum_{j=1}^L G_{\tau}(e_j(l)) e_j(l) \frac{\partial e_j(l)}{\partial \mathbf{x}(l|l)} = 0 \quad (59)$$

The above equation is rewritten in the following manner:

$$\left(\frac{\partial W(l)}{\partial \mathbf{x}} \right)^T C(l) [D(l) - W(l)] = 0 \quad (60)$$

where, $C(l) = \begin{bmatrix} C_{x,l} & 0 \\ 0 & C_{y,l} \end{bmatrix}$, and $C_{x,l}, C_{y,l}$ are defined as follows:

$$C_{x,l} = \operatorname{diag}(G_{\tau}(e_1(l)), \dots, G_{\tau}(e_n(l))) \quad (61)$$

$$C_{y,l} = \operatorname{diag}(G_{\tau}(e_{n+1}(l)), \dots, G_{\tau}(e_{n+m}(l))) \quad (62)$$

The measurement information is recorrected via $C(l)$. Define the new covariance matrix as follows:

$$\begin{aligned} \Lambda(l) &= B(l)C^{-1}(l)B^T(l) \\ &= \begin{bmatrix} B_p(l|l-1)C_{x,l}^{-1}B_p^T(l|l-1) & 0 \\ 0 & B_r(l)C_{y,l}^{-1}B_r^T(l) \end{bmatrix} \end{aligned} \quad (63)$$

Thus, the corrected $P^*(l|l-1)$ and the corrected R^*_l can be obtained.

$$\dot{P}(l|l-1) = B_p(l|l-1)C_{x,l}^{-1}B_p^T(l|l-1) \quad (64)$$

$$\dot{R}_l = B_r(l)C_{y,l}^{-1}B_r^T(l) \quad (65)$$

Then, recalculating the Kalman Gain K^* and update error estimate.

$$\dot{K} = \dot{P}(l|l-1)H^T(H\dot{P}(l|l-1)H^T + \dot{R}_l)^{-1} \quad (66)$$

$$\delta\hat{\mathbf{x}}(l|l) = \delta\hat{\mathbf{x}}(l|l-1) + \dot{K}(\mathbf{z}_n(l) - \mathbf{z}(l|l-1)) \quad (67)$$

Finally, the updated error state vector is employed to rectify the INS state estimate $\hat{\mathbf{x}}^*(l)$, thereby fusing the output tag state estimate.

$$\hat{\mathbf{x}}(l|l) = \hat{\mathbf{x}}(l) - \delta\hat{\mathbf{x}}(l|l) \quad (68)$$

7. Simulation and experimental results

During the experiment, NaveGo is utilized as a simulation tool for performing simulation verification to assess the efficacy and robustness of the AMCKKF fused localization algorithm. In the simulation experiments, we assume that the LOS errors follow a Gaussian distribution, while the NLOS errors are modeled using both gamma and Gaussian distributions. Subsequently, real experiments were conducted in a practical environment to further validate the performance of the algorithms. The following algorithms were selected for comparison: EKF, Sage-Husa Fussy Adaptive Filter (SHFAF) [26], Interacting Multiple Model-Maximum Correntropy Kalman Filter (IMM-MCKF) [27], and Modified Verification Gate Generalized Probability Data Association Filter (MVG-GPDF) [28].

7.1. Simulation scheme

In this paper, we consider localization within a horizontal area of 60 m × 60 m. The tag move within this area, while the four BSs are randomly deployed within the localization region. In the simulation, the height of the tag is kept constant, with the z-coordinate is fixed at 2 m, as we are only concerned with the localization accuracy in the 2D plane. The tag node moves within this region, with no consideration for changes in vertical direction, resulting in the absence of roll, pitch, and vertical acceleration. The Root Mean Square Error (RMSE) and Cumulative Distribution Function (CDF) are employed as the criteria to evaluate the localization accuracy of distinct algorithms.

$$RMSE = \sqrt{\frac{1}{T_{MC} \times L} \sum_{i=1}^{T_{MC}} \sum_{l=1}^L ||\mathbf{x}_l(7:9) - \hat{\mathbf{x}}_{l/l}(7:9)||^2} \quad (69)$$

where, $L=16282$ is the overall count of samples, T_{MC} denotes the Monte Carlo iterations. $\mathbf{x}_l(7:9)$ denotes the tag's true coordinates at the l -th time, while $\hat{\mathbf{x}}_{l/l}(7:9)$ denotes the estimated coordinates at the l -th time. The specific settings of the INS device's filter are presented in Table 1.

7.2. The NLOS error follows a gamma distribution

The NLOS noise model, which is designed to reflect the characteristics of indoor environments, is assumed to be a Gamma distribution. The expression for this is as follows:

Table 1
INS parameter setting.

IMU	Parameter Name	Value
Accelerometer	Random Walk (ARW)	$0.02\text{m}/(\text{s}\cdot\sqrt{\text{h}})$
	Bias	6 mg
	Correlation time	100 s
	Frequency	100 Hz
	Random Walk (VRW)	$0.125^\circ/\sqrt{\text{h}}$
Gyroscope	Bias	$0.2^\circ/\text{s}$
	Correlation time	100 s
	Frequency	100 Hz

$$f(d_{\text{NLOS}}) = \beta \frac{(\beta d_{\text{NLOS}})^{\alpha-1}}{\Gamma(\alpha)} e^{(-d_{\text{NLOS}}/\beta)} \quad (70)$$

where, d_{NLOS} represent the error caused by NLOS transmission, α denotes the shape parameter and β denotes the scale parameter. The mean value of d_{NLOS} and variance of d_{NLOS} , which respectively equal $\frac{\alpha}{\beta}$, $\frac{\alpha}{\beta^2}$.

The noise measurement LOS is presumed to adheres to the Gaussian distribution with the mean $u = 0$ and the standard deviation $\sigma_{\text{LOS}} = 0.1$. In the event that the NLOS error conforms to the gamma distribution, the pertinent filter parameters for UWB devices are delineated in Table 2.

According to the parameters in the above table, the RMSE and CDF of several algorithms for 1000 Monte Carlo simulation experiments are depicted in the figures below.

As illustrated in Fig. 3, the proposed algorithm consistently achieves significantly lower RMSE values than other comparative algorithms across different time intervals. The average RMSE associated with the AMCCKF is 0.2706 m, whereas EKF, SHFAF, IMM-MCKF, and MVG-GPDAF exhibit average errors of 1.24 m, 1.32 m, 0.593 m, and 0.78 m, respectively. Compared to EKF, SHFAF, IMM-MCKF, and MVG-GPDAF, our algorithm improves accuracy by 78.19 %, 79.50 %, 54.37 %, and 65.33 %, respectively.

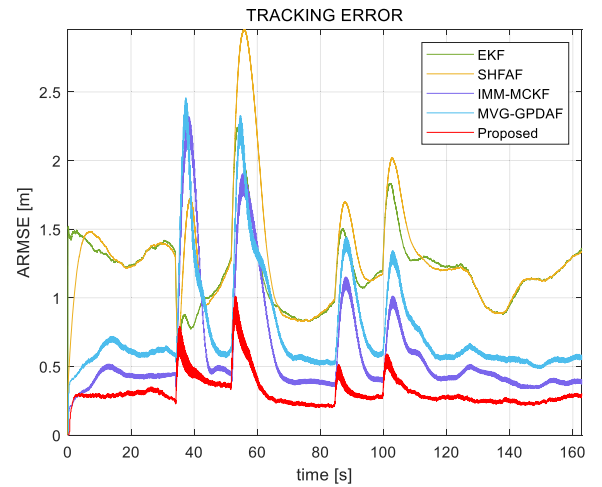
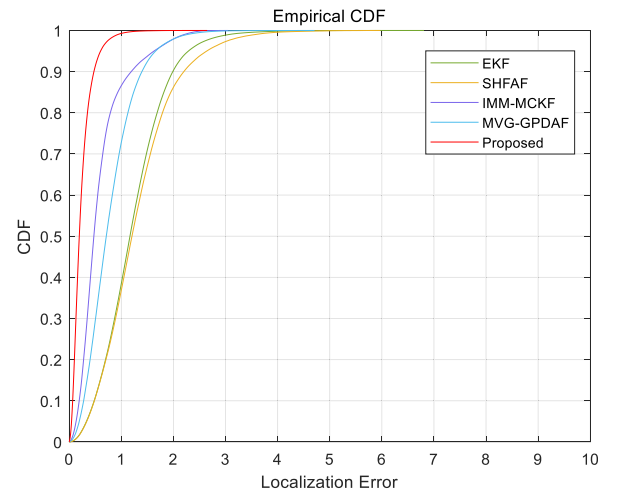
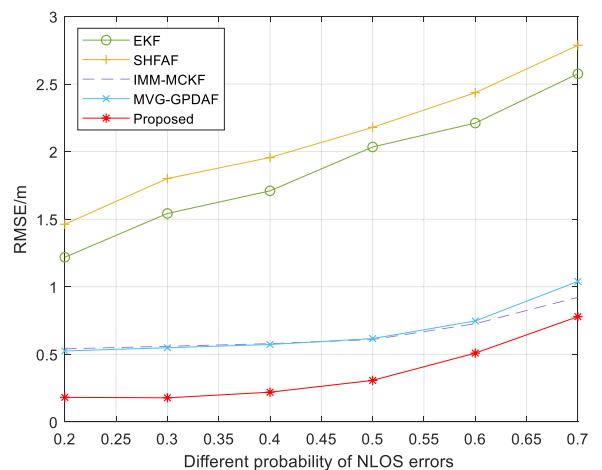
Fig. 4 shows the CDF, demonstrating the superior stability of the AMCCKF in comparison to the other four methods. The AMCCKF achieves 90 % of localization errors under 2.211 m, whereas EKF, SHFAF, IMM-MCKF, and MVG-GPDAF have errors under 3.870 m, 4.102 m, 3.667 m, and 2.692 m, respectively. At a cumulative error probability close to 1, the AMCCKF's localization error is only 1.462 m.

Fig. 5 demonstrates the effect of varying NLOS error probabilities upon the RMSE. As the probability P_{NLOS} rises from 0.2 to 0.7, the RMSE values for EKF, SHFAF, IMM-MCKF, MVG-GPDA, and the proposed algorithm are 1.882 m, 2.104 m, 0.6564 m, 0.6751 m, and 0.3627 m, respectively. This corresponds to accuracy improvements of 78.19 %, 79.50 %, 54.37 %, and 65.63 %, respectively. Consequently, the AMCCKF algorithm demonstrates superior capability in mitigating NLOS interference.

Fig. 6 presents the RMSE outcomes for several algorithm as the mean μ gradually increases. While the RMSE of algorithms such as EKF, SHFAF, and IMM-MCKF correspondingly increases, MVG-GPDAF initially experiences an increase followed by a decrease. This behavior is attributed to the significant noise intensity, leading to data

Table 2
UWB parameter setting.

Parameter Name	Symbol	Value
Number of base stations	M	5
Probability of NLOS error occurrence	P_{NLOS}	0.5
LOS noise	$N(u, \sigma_{\text{LOS}}^2)$	$N(0, 0.1^2)$
NLOS noise	$N(u_{\text{NLOS}}, \sigma_{\text{NLOS}}^2)$	$N(3, 1.2)$
Gamma factor	(α, β)	(36, 12)
Frequency	f	50 Hz

**Fig. 3.** The average RMSE versus time.**Fig. 4.** CDF for localization errors.**Fig. 5.** Variation of RMSE with varying probabilities of NLOS errors P_{NLOS} .

elimination. In contrast, our proposed algorithm exhibits slower RMSE growth and consistently maintains superior localization accuracy compared to other methods, indicating superior localization performance.

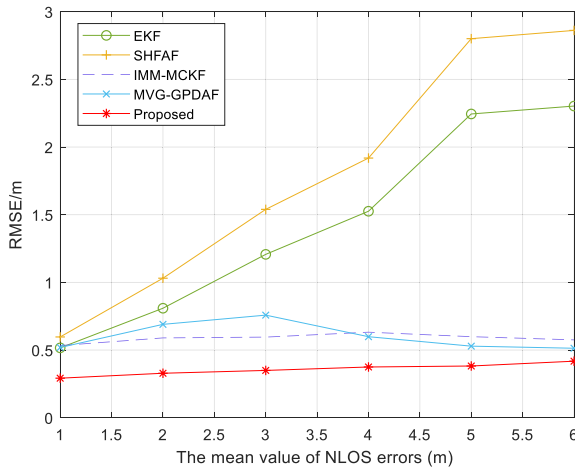


Fig. 6. Variation of RMSE with varying mean uof NLOS noise.

Fig. 7 illustrates that with the increase in the standard deviation σ , the RMSE values for EKF, SHFAF, IMM-MCKF, and the proposed algorithm all gradually increases, whereas the RMSE of MVG-GPDAF shows a slight decrease. This behavior can be attributed to the effectiveness of clustering algorithms in eliminating outlier data when significant differences exist between clusters. The RMSE of EKF, SHFAF, IMM-MCKF, MVG-GPDAF, and AMCKKF is 1.292 m, 1.439 m, 0.639 m, 0.675 m, and 0.4099 m, respectively. Compared to EKF, SHFAF, and IMM-MCKF, the AMCKKF achieves localization accuracy improvements of 68.30 %, 71.54 %, 35.88 %, and 39.28 %, respectively.

7.3. The NLOS error follows a Gaussian distribution

To verify the proposed algorithm's performance in comparison to other algorithms, we assume that the indoor noise conforms to a Gaussian distribution. Table 3 lists the simulation parameters.

According to the parameters in the above table, the RMSE and CDF of several algorithms for 1000 Monte Carlo simulation experiments are depicted in Figs. 8 and 9, respectively.

Fig. 8 illustrates that the AMCKKF outperforms other methods with regard to localization precision, particularly during changes in motion states where peak errors are significantly smaller. The proposed algorithm achieves RMSE of 0.4676 m, while EKF, SHFAF, IMM-MCKF, and MVG-GPDAF exhibit average errors of 1.645 m, 1.713 m, 0.6975 m, and 0.6076 m, respectively. These improvements correspond to localization accuracy enhancements of 71.54 %, 72.63 %, 32.94 %, and 23.05 %, respectively.

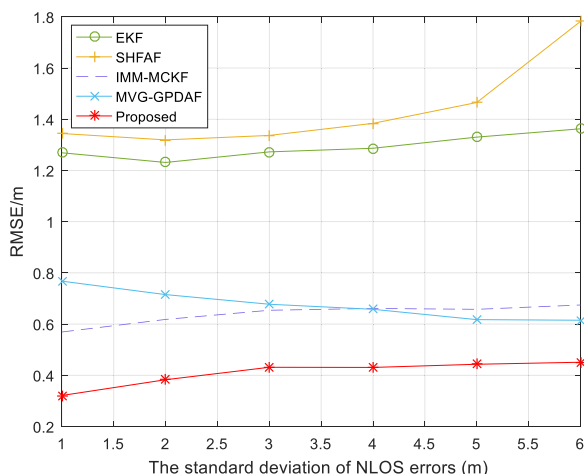


Fig. 7. Variation of RMSE with standard deviation σ of NLOS errors.

Table 3

UWB parameter setting.

Parameter Name	Symbol	Value
Number of base stations	M	5
Probability of NLOS error occurrence	P_{NLOS}	0.5
LOS noise	$N(u, \sigma_{LOS}^2)$	$N(0, 0.1^2)$
NLOS noise	$N(u_{NLOS}, \sigma_{NLOS}^2)$	$N(4, 6)$
Frequency	f	50 Hz

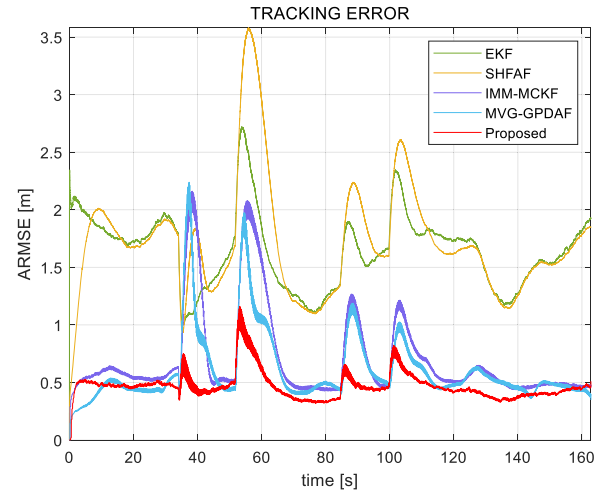


Fig. 8. The average RMSE versus time.

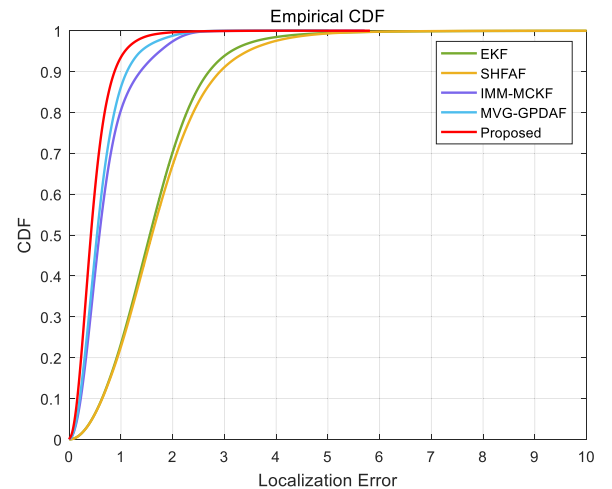


Fig. 9. CDF for localization errors.

respectively, compared to the aforementioned algorithms.

Fig. 9 presents the CDF plot of localization errors using default parameter settings. Upon reaching a cumulative probability of 90 %, the positioning errors for the AMCKKF is 0.91 m, while the errors for EKF, SHFAF, IMM-MCKF, and MVG-GPDAF are 1.117 m, 1.423 m, 2.704 m, and 2.91 m, respectively. The AMCKKF method exhibits superior stability compared to the other methods.

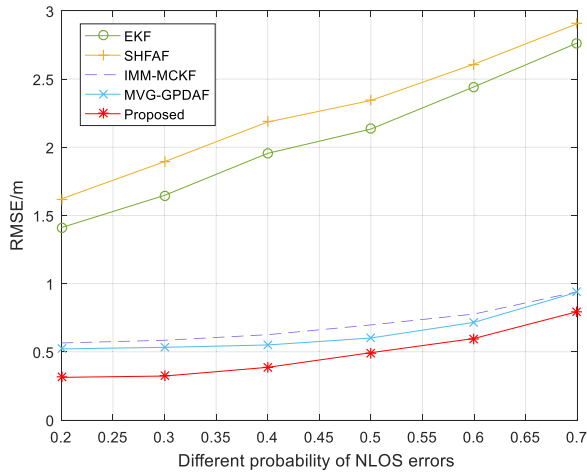
In this simulation environment, we compared the computational complexity of these algorithms, recording their average runtime per iteration and positioning accuracy, as presented in Table 4. The results indicate that the processing time per iteration remains below the sampling interval (UWB sampling frequency of 50 Hz), thereby satisfying the requirements for real-time tracking.

As shown in Fig. 10, when the P_{NLOS} rises to 0.7, the RMSE values for

Table 4

Comparison of total running time and error.

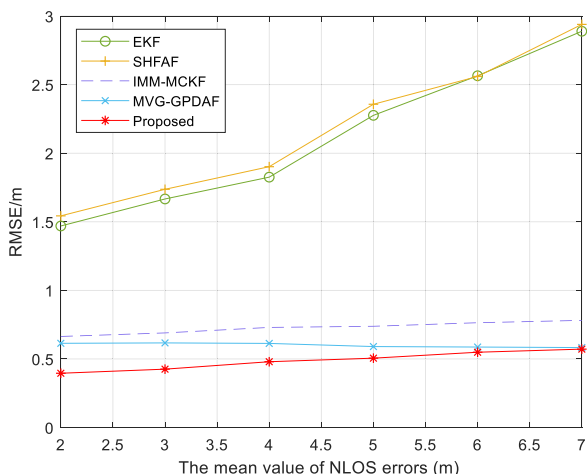
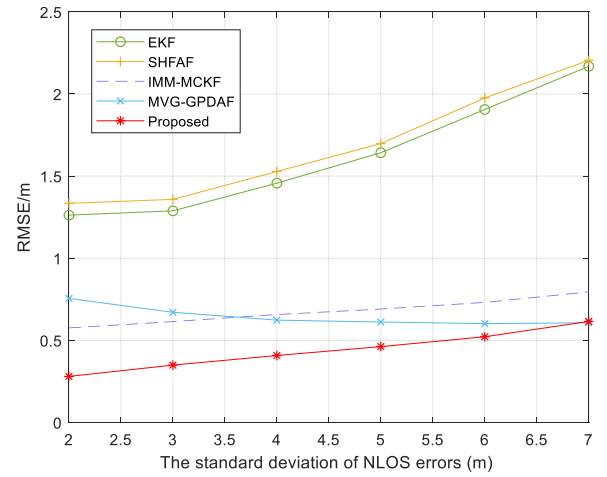
Algorithm	Running Time	Positioning Error
EKF	0.003453 s	1.645 m
SHFAF	0.003015 s	1.713 m
IMM-MCKF	0.001506 s	0.6975 m
MVG-GPDAF	0.001321 s	0.6076 m
AMCCKF	0.001621 s	0.4676 m

**Fig. 10.** Variation of RMSE with varying probabilities of NLOS errors P_{NLOS} .

EKF, SHFAF, IMM-MCKF, MVG-GPDA and the proposed algorithm are 2.059 m, 2.26 m, 0.6982 m, 0.6432 m, 0.4845 m, respectively. Compared to the other algorithms, this corresponds to accuracy improvements of 76.47 %, 78.57 %, 30.61 %, and 24.68 %. The proposed algorithm also effectively reduces NLOS interference in Gaussian environments.

Fig. 11 illustrates the RMSE results for various algorithms as the mean μ gradually increases. Specifically, the RMSE values for EKF, SHFAF, and IMM-MCKF increase correspondingly, while MVG-GPDAF exhibits a gradual reduction. In contrast, the AMCCKF consistently maintains better positioning precision compared to other algorithms, demonstrating its superior localization performance.

Fig. 12 shows that with the increase in the standard deviation σ , the RMSE values for EKF, SHFAF, IMM-MCKF, and the proposed algorithm gradually increases, while the RMSE value for MVG-GPDAF gradually decreases. This behavior is attributed to the MVG-GPDAF algorithm

**Fig. 11.** Variation of RMSE with varying mean value u of NLOS errors.**Fig. 12.** Variation of RMSE with standard deviation σ of NLOS errors.

effectively excluding data with large errors. Specifically, the RMSE values for EKF, SHFAF, IMM-MCKF, MVG-GPDAF, and the AMCCKF are 1.62 m, 1.683 m, 0.6771 m, 0.6447 m, and 0.4393 m, respectively. Compared to EKF, SHFAF, and IMM-MCKF, the AMCCKF improved the positioning accuracy by 72.85 %, 73.89 %, 35.12 %, and 31.87 %, respectively.

7.4. Experiment

To further validate the efficacy of the AMCCKF algorithm in practical settings. We carried out validation tests within an actual indoor scenario of 10 m*14 m. The tag node was fixed to move on an inertial navigation trolley to achieve time synchronization between the UWB and INS modules. Before the start of the experiment, we preset the trolley's forward path and recorded the positional coordinate data. During the experiment, the moving trolley followed the preset path, resulting in the pre-experimental data aligning with the data gathered during the experiment. The inertial navigation returns data at a frequency of 50 Hz. In order to calculate the distance from the tag node to the BSs, we use the TOA technique for ranging. the INS measurements are acquired by the IMU vehicle. The UWB and INS devices are shown in Fig. 13.

During the experiment, four sites were selected as base stations for UWB distance measurement. The trolley is traveling along a pre-determined route at a speed of approximately 0.5 m/s, with non-uniform motion around corners. The tag trolley moves along a pre-defined route, and the trajectory, direction, and location of the base station are shown in Fig. 14.

We evaluated the localization accuracy of the algorithms using RMSE

**Fig. 13.** INS and UWB device.

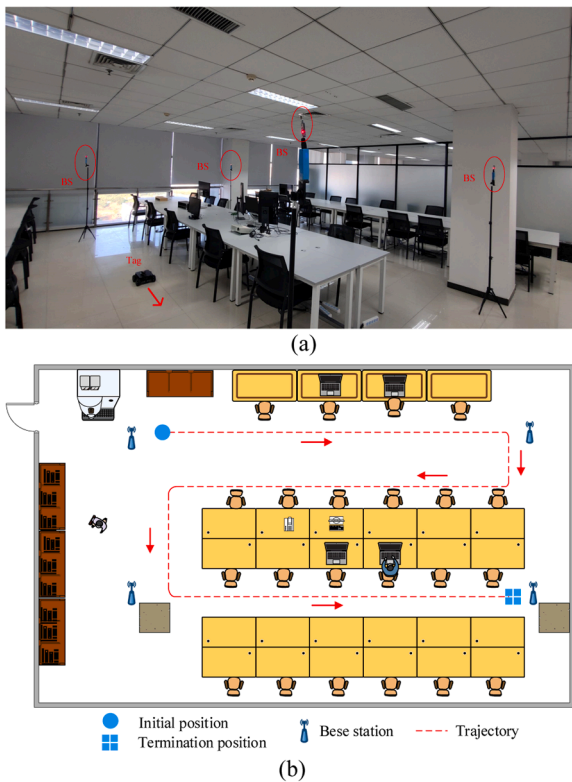


Fig. 14. (a) Real scene of the offices (b) Plan of offices.

and CDF plots for each measurement point.

Fig. 15 demonstrates that the AMCKF consistently achieves lower error values than other methods, with most errors remaining within 0.5 m. The average localization errors for EKF, SHFAF, IMM-MCKF, and MVG-GPDA are 1.182 m, 1.259 m, 0.5424 m, and 0.7659 m, respectively, while AMCKF attains an average error of only 0.2662 m. This demonstrates that the AMCKF algorithm is significantly more effective in handling NLOS conditions, outperforming the other four algorithms.

As shown in Fig. 16, the CDF reveals the following 90 % localization errors for various algorithms: The errors for EKF, SHFAF, IMM-MCKF, and MVG-GPDAF are 1.726, 1.93 m, 0.8738 m, and 1.108 m, respectively, while AMCKF attains an error of only 0.5154 m. This indicates that, in comparison to the other four methods, the AMCKF exhibits higher stability.

8. Conclusion and limitations

This paper presents an enhanced AMCKF with adaptive behavior for integrating INS/UWB positioning algorithms. Taking advantage of both INS and UWB technologies, it combines maximum correlation entropy to reduce the effect of non-Gaussian noise and introduces a novel PSO-kmeans clustering method to reduce NLOS noise in UWB. For complex environments where the noise characteristics change over time. We propose an enhanced adaptive Sage-Husa filtering algorithm that utilizes previous process estimation and recording information to suppress the effect of outliers, and modifies the expression of the covariance matrix and ensures its positive qualitative nature, which effectively improves the adaptability and stability of localization. Finally, We propose a novel adaptive bandwidth selection method based on NLOS detection. It solves the problem of over-convergence caused by outliers, achieves adaptation to specific measurements, and ensures optimal estimation in a mild NLOS environment. Comparative results show improved adaptation and robustness to non-Gaussian noise with higher localization accuracy compared to recent methods, and the results demonstrate its effectiveness in practical applications. Future research

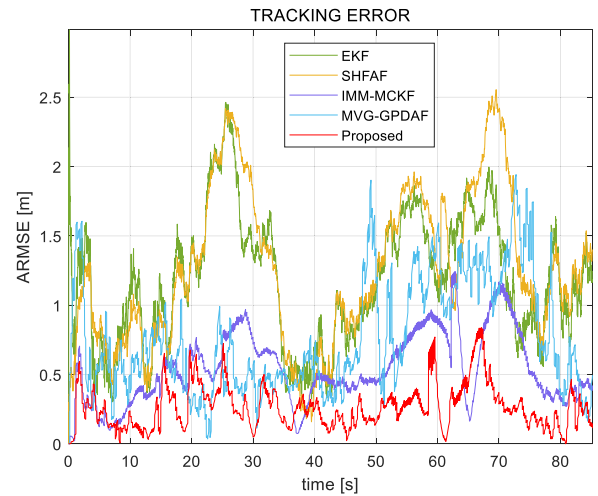


Fig. 15. The RMSE at each point over time.

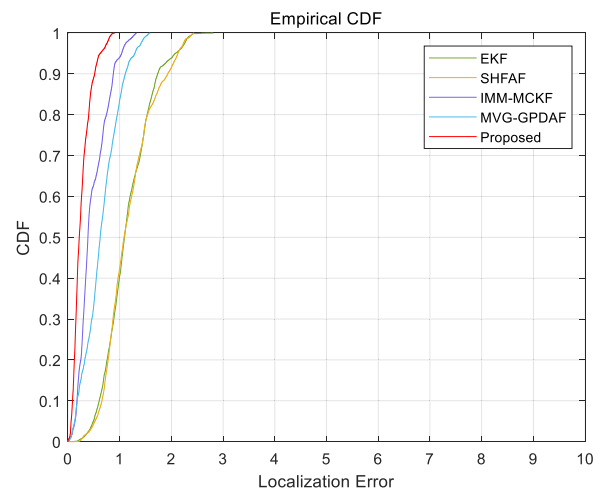


Fig. 16. CDF for localization error at each point.

will encompass the extension to multi-sensor systems and multi-target localization.

CRediT authorship contribution statement

Yan Wang: Methodology, Investigation. **Shengqing Fu:** Writing – original draft, Software. **Fuhui Wang:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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